

High Performance Graphics 2024 - HPG 2024

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ABSTRACT

This panel will present a short overview of the High-Performance Graphics (HPG) 2024 conference program. The conference is co-sponsored by ACM SIGGRAPH and Eurographics and, in 2024, is co-located with SIGGRAPH, taking place during the previous days, in Denver. HPG brings together researchers, engineers, and architects to discuss the complex interactions of parallel and custom hardware, novel programming models, and efficient algorithms in the design of systems for current and future graphics and visual computing applications. The panel will comprise short presentations by the keynote speakers, who will highlight their talks, and by the program and paper chairs, who will present a short overview of the papers program. This panel is an opportunity for all SIGGRAPH attendants to have an overview of the HPG content. Since it happens before SIGGRAPH starts, we understand that taking a short overview of the conference content can be of great value for both conferences.

KEYWORDS

Real-Time, Rendering, Hardware, AI

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1 INTRODUCTION

High-Performance Graphics (HPG), co-sponsored by ACM SIGGRAPH and EUROGRAPHICS, is the leading international forum for performance-oriented graphics and imaging systems research, including innovative algorithms, efficient implementations, languages, compilers, parallelism, and hardware architectures for high-performance graphics. The conference brings together researchers, engineers, and architects to discuss the complex interactions of parallel and custom hardware, novel programming models, and

efficient algorithms in the design of systems for current and future graphics and visual computing applications. HPG 2024 will be co-located with SIGGRAPH in Denver, Colorado on Friday, July 26, through Sunday, July 28. The main topics of interest to HPG are strongly connected with SIGGRAPH and include dedicated hardware, Machine learning techniques for high-performance graphics; Neural Rendering, Real-time and interactive ray tracing hardware or software; Large-Scale, Parallel and Cloud Applications in Computer Graphics; High-Performance applications in film, visual effects, professional and scientific visualization, among other topics. The panel brings to SIGGRAPH participants an overview of the HPG 2024 conference content.

The central question for the panelists will be “What do the trends from HPG 2024’s Program mean for the future of high-performance graphics?”. Through that organizing question, we will present an overview of the topics to give panel attendees a flavor of the HPG 2024 content, and then offer new and interesting views of what that means going forward. Moreover, we have industry practitioners and academic researchers on the panel, each of whom will provide diverse, distinct, and perhaps conflicting views of what those trends from HPG 2024 actually are and what they mean for the near- and long-term future of high-performance graphics as a field. One of HPG’s strengths is that it has extensive participation from both academic and industrial researchers, as well as financial contributions from industry. This year, HPG is sponsored by Reality Labs, Activision, Adobe, AMD, Electronic Arts, NVIDIA, Samsung, Qualcomm, and the Khronos Group. The conference theme in 2024 is Machine Learning in High Performance Graphics. HPG receives full paper contributions, posters, and Hot3D presentations, where companies present their latest and greatest graphics hardware and software products. The panelist will be composed by the following speakers:

- Esteban Clua (HPG 2024 Program chair). He is full professor at Universidade federal fluminense, Brazil. He is the director of the UFF Medialab at the Computer Science Institute. Esteban is one of the founders of the Brazilian Symposium of Games and Digital Entertainment.
- John Burgess (keynote speakers at HPG 2024). Vice President of GPU Architecture at NVIDIA, where he has been contributing to GPU design for 20 years. His experience spans memory system design to streaming processor cores, including tensor cores for AI and Deep Learning and ray tracing cores for professional visualization and gaming.
- Karen Ghavam (keynote speakers at HPG 2024). After a successful career at Hewlett Packard for 35 years, Karen

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decided to “retire” from Hewlett Packard to take the helm at LunarG as CEO and Engineering Director. LunarG is a graphics software consultancy that delivers many of the Vulkan ecosystem developers tools for the Vulkan API. Karen has been driving the Vulkan Ecosystem Developer tools since before the launch of the Vulkan API in February of 2016.

- Peter Shirley (keynote speaker at HPG 2024) is a Vice President of Computer Graphics at Activision. He is a former researcher at NVIDIA, a co-founder two software companies, and a Professor at Indiana University, Cornell University, and the University of Utah. He is the coauthor of several books on computer graphics and a variety of technical articles.
- Christiaan Gribble (General Chair at HPG 2024). He is a PMTS Silicon Design Engineer at Advanced Micro Devices,

Inc. (AMD), where he is helping to design the next generation of real-time ray tracing (RTRT) hardware architectures. His research explores the synthesis of RTRT, large-scale modeling and simulation systems, and hardware architecture. Previously, Gribble was the Director of High Performance Computing (HPC).

- Won-Jong Lee (HPG Hot3D chair) is a principal GPU architect at AMD, designing the core blocks of ray tracing hardware of next-generation GPUs. Previously, he was a research scientist at Intel, where he led several research projects aiming at Intel’s Xe GPUs such as a new programming model and visibility-driven BVH build.

HPG 2024 – <https://www.highperformancegraphics.org/2024>.